

Temporal trends in neighborhood access to daily destinations across 12 Spanish cities (2014–2024)

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MAIN FINDING

Poster Number
I-P-H.01

Daily living access improved in all 12 Spanish cities over a decade.
Low-access areas improved fastest: convergence dominates, although heterogeneity in pace.
Yet stagnant pockets persist, warranting targeted policy.

12

Spanish cities
surveyed annually 2014–2024

98%

grid cells
with complete 11-year series

3

objective indicators within 500 m
food market, convenience store, public transport

+0.38 SD

mean change / decade

Background

Public policies are increasingly investing in the promotion of creating more livable and walkable environments. The purpose of this study was to **examine temporal trends in neighborhood access to daily destinations across multiple Spanish cities between 2014 and 2024.**

Specifically, we aimed to **quantify how mean access to daily destinations evolved over time within and between cities**, identifying whether urban environments became more or less supportive of walking during this period.

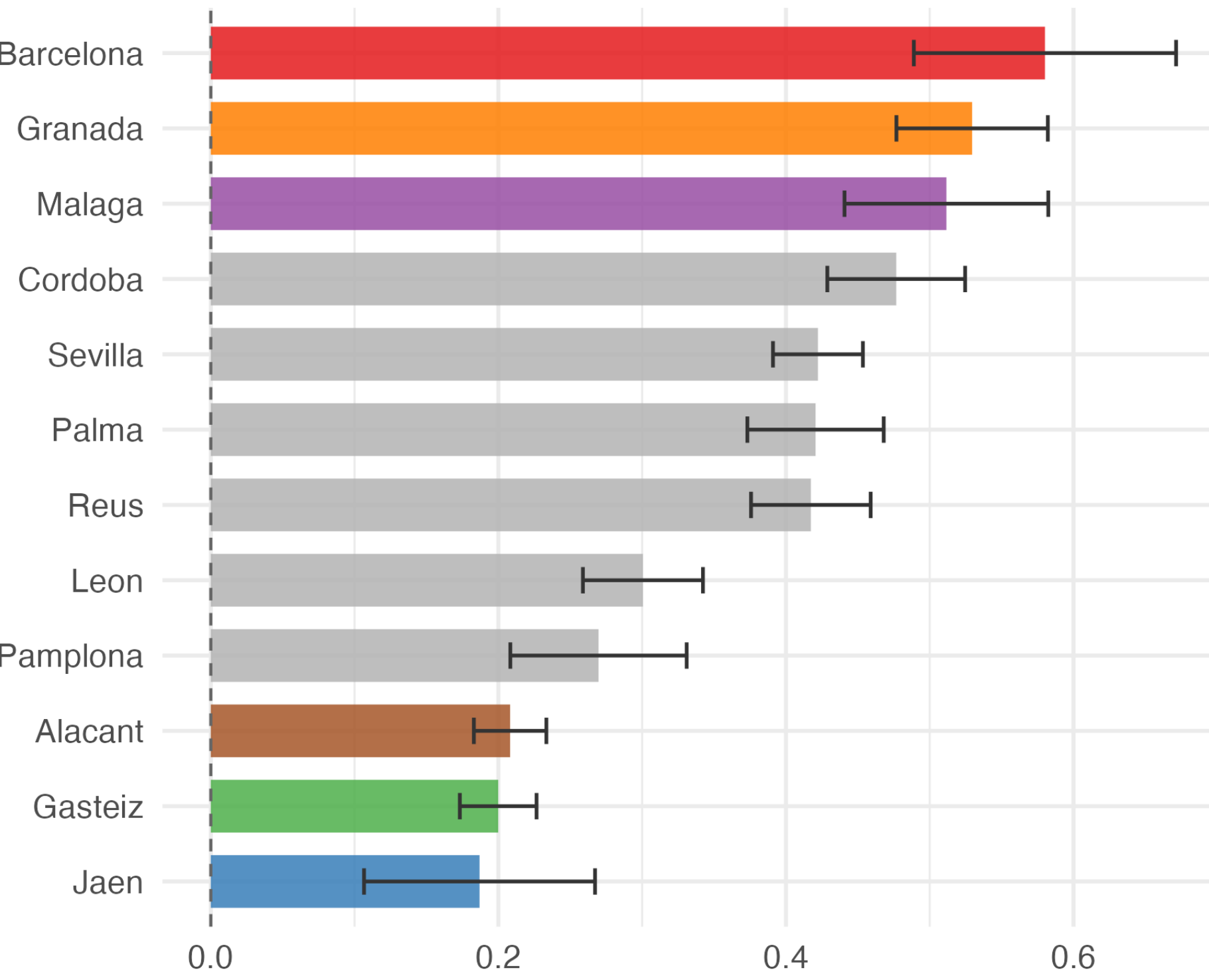
Methods

We conducted a **longitudinal ecological analysis of neighborhood access to daily destinations across 12 geographically diverse Spanish cities from 2014 to 2024.**

Access to daily destinations was quantified annually within 100 × 100 m grid cells following the **Global Healthy and Sustainable City Indicators Software**, an open science toolkit. A **composite daily living index** was calculated summing equal-weighted access scores on three objective indicators derived from geospatial data at the neighborhood level: **healthy food markets, convenience stores, and public transport stops** within 500 m.

To enable temporal and intercity comparisons, **daily living index values were standardized as z-scores using the global mean and standard deviation of all observations across cities and years.** For each city and year, we computed the mean daily living index and its corresponding 95% confidence interval (95% CI).

Net change in the daily living index by city over the study period (2014–2024)

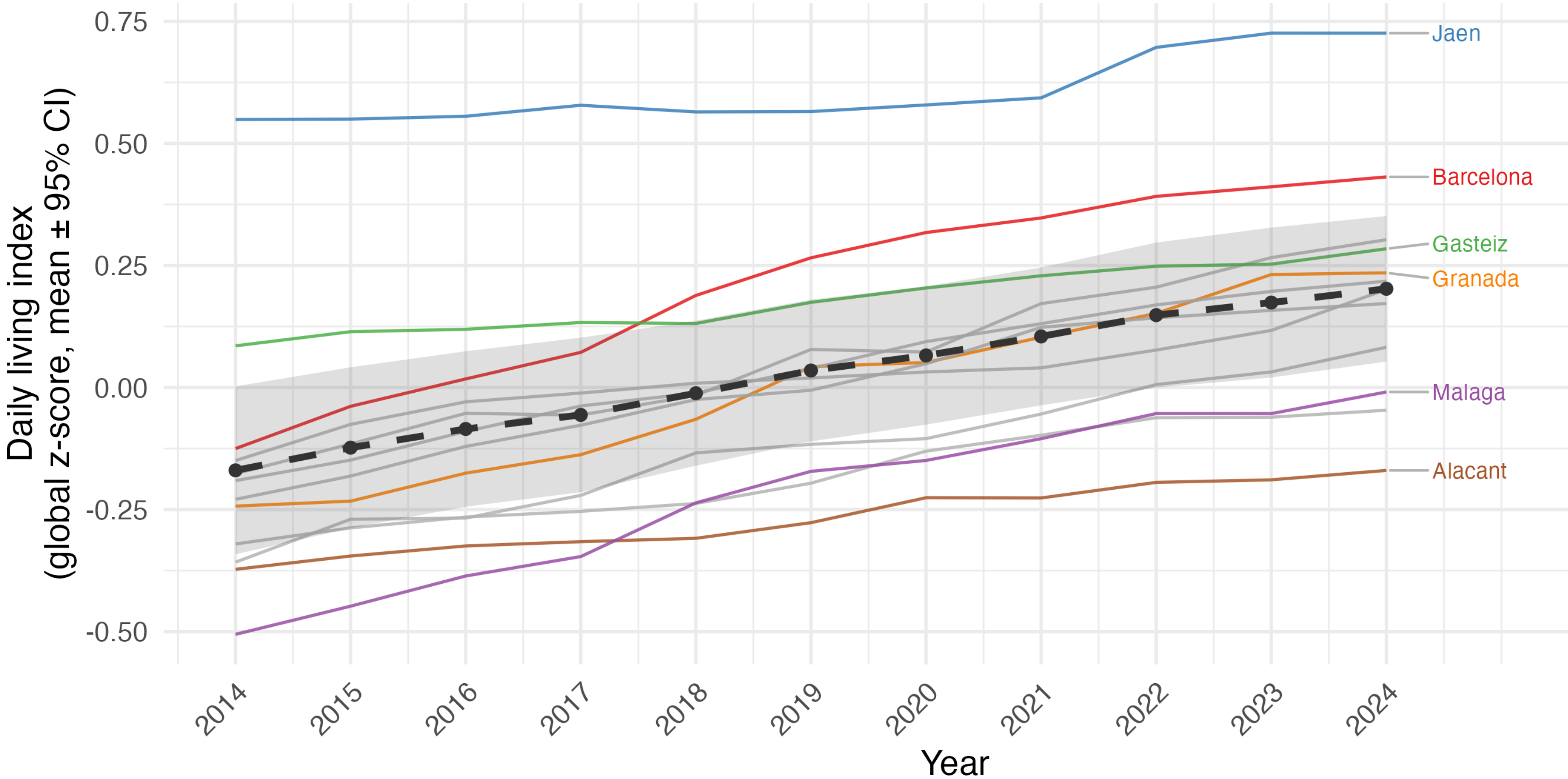


Mean change in the standardized daily living index (SD units) per city. Bars distinguish cities with sustained improvement from those showing relative stagnation.

Conclusions

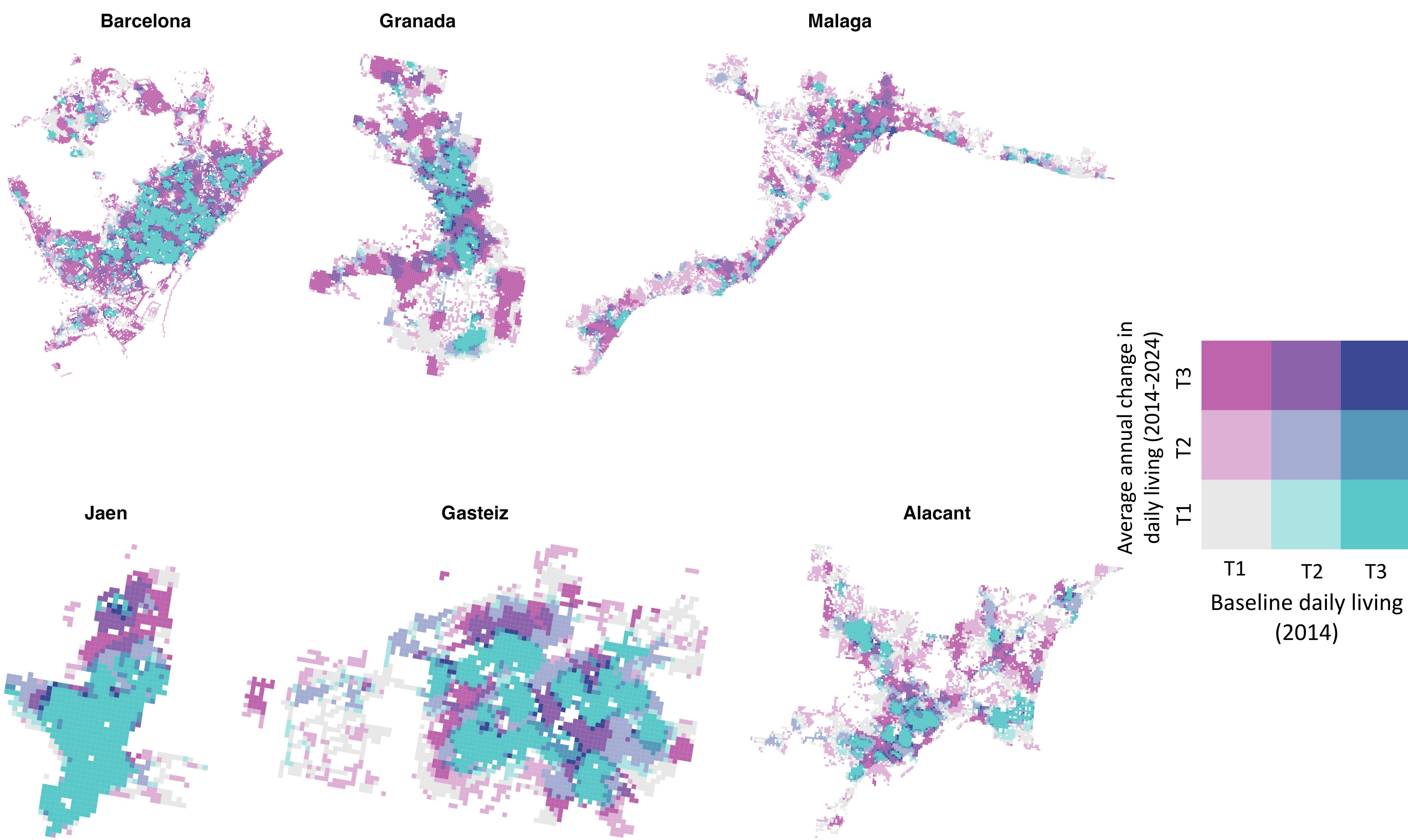
This study provides a **standardized, spatially and temporally comparable framework to monitor changes in neighborhood access** to daily destinations across Spanish cities over the past decade. The resulting longitudinal access to daily destinations dataset establishes the **foundation for subsequent investigations linking environmental dynamics with behavioral and health outcomes**, including analyses of how the lifestyle intervention is influenced by neighborhood context.

Annual trends in the daily living index across 12 Spanish cities (2014–2024)



Annual mean standardized daily living index (z-score) per city with 95% confidence intervals. Each line represents one city; shaded bands indicate interannual variability. Warm colors represents faster-growing cities (Barcelona, Granada, Málaga) and cool colors represents slowest-growing cities (Alacant, Gasteiz, Jaén).

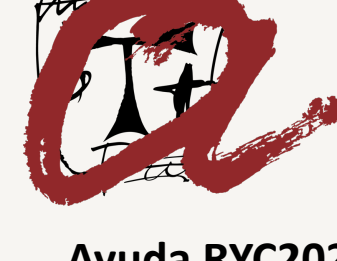
Spatial distribution of the daily living index at the 100 × 100 m grid-cell level



Cartographic representation of the standardized access index across the 6 cities, 3 faster-growing cities (Barcelona, Granada, Málaga) and 3 slowest-growing cities (Alacant, Gasteiz, Jaén). x-axis layout represents baseline daily living z-score (2014) y-axis represents annual slope (z-score/year). Grid cells 100 x 100 m. n = 71.928 cells with complete 11-years series.



Proyectos código PI25/01538, PI25/01521



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interactive
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